Vim Editor Notes

🔹 Introduction to Vim

Vim (Vi Improved) is a powerful text editor that comes pre-installed in most Linux and UNIX systems. It is an enhanced version of the Vi editor and is widely used by programmers for its speed, efficiency, and keyboard-based editing.

Vim is a **modal editor**, meaning it has different modes for editing, navigating, and executing commands.

🔹 Modes in Vim

1. **Normal Mode (Command Mode)**
   - Default mode after opening Vim.
   - Used for navigation, deleting, copying, and pasting text.
   - Press Esc to return to this mode from any other mode.
2. **Insert Mode**
   - Used for inserting or editing text.
   - Commands to enter Insert Mode:
     - i → Insert before cursor
     - a → Insert after cursor
     - o → Open a new line below
     - O → Open a new line above
3. **Visual Mode**
   - Used to select text.
   - Enter using v (character-wise) or V (line-wise).
4. **Command-Line Mode (Ex Mode)**
   - Used to enter commands that start with a colon (:).
   - Example: :wq to save and exit.

🔹 Basic File Commands

Action	Command
Open a file	vim filename
Save file	:w

Action	Command
Quit Vim	:q
Save and quit	:wq or :x
Quit without saving	:q!
Save as another name	:w newfile

🔹 Cursor Movement Commands

Action	Command
Left	h
Right	l
Up	k
Down	j
Beginning of line	0
End of line	$
Top of file	gg
Bottom of file	G
Go to line number	:n

🔹 Text Editing Commands

Action	Command
Delete character	x
Delete line	dd
Copy (yank) line	yy
Paste below cursor	p
Paste above cursor	P
Undo last change	u
Redo change	Ctrl + r
Replace single character	r + character

Action	Command
Join two lines	J

🔹 Searching and Replacing

Action	Command
Search for word	/word
Search backward	?word
Next search result	n
Previous search result	N
Replace a word	:%s/old/new/g
Confirm each replacement	:%s/old/new/gc

🔹 Visual Mode Operations

Action	Command
Enter visual mode	v
Enter visual line mode	V
Copy selected text	y
Cut selected text	d
Paste selected text	p

🔹 Working with Multiple Files

Action	Command
Open multiple files	vim file1 file2
Switch to next file	:n
Switch to previous file	:N
List open files	:ls
Switch to specific file	:b n (where n is buffer number)

🔹 Useful Settings

Setting	Command
Enable line numbers	:set number
Disable line numbers	:set nonumber
Enable syntax highlighting	:syntax on
Disable syntax highlighting	:syntax off
Enable auto indentation	:set autoindent
Enable search highlight	:set hlsearch
Disable search highlight	:set nohlsearch

🔹 Exiting Vim

Action	Command
Quit without saving	:q!
Save and quit	:wq
Save file	:w
Quit all files	:qa
Save all and quit	:wqa

🔹 Tips for Beginners

- Press Esc frequently to ensure you're in **Normal Mode**.
- Combine commands with numbers for efficiency (e.g., 3dd deletes 3 lines).
- Use :help to open Vim's built-in documentation.
- Customize Vim using the **.vimrc** configuration file.

🔹 Summary

- **Vim** is a highly efficient text editor used mainly by programmers.
- It operates in **multiple modes** to separate text editing and command execution.
- Mastering Vim's shortcuts increases productivity and editing speed.